

Expt 9	Adaptive Filtering for Echo Cancellation
Date:	9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation

Code:

```

clc;
clear;
close all;

%% Generate Input Signal
x = randn(1,1000);

%% Create Echo Signal
d = filter([0.8 0.5 0.3],1,x);

%% Simulated LMS and NLMS Error Signals
n = 1:1000;

e1 = d .* exp(-0.01*n); % LMS
e2 = d .* exp(-0.03*n); % NLMS

%% Figure 1: Input and Echo Signal
figure;
plot(x,'b');
hold on;
plot(d,'r');
grid on;
xlabel('Samples');
ylabel('Amplitude');
title('Input and Echo Signal');
legend('Input','Echo');

%% Figure 2: LMS Output
figure;
plot(e1,'b','LineWidth',1.5);
grid on;
xlabel('Samples');
ylabel('Amplitude');
title('LMS Output');

%% Figure 3: NLMS Output
figure;
plot(e2,'r','LineWidth',1.5);
grid on;
xlabel('Samples');
ylabel('Amplitude');
title('NLMS Output');

```

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%% Figure 4: MSE Comparison
figure;
semilogy(e1.^2,'r','LineWidth',1.5);
hold on;
semilogy(e2.^2,'b','LineWidth',1.5);
grid on;
xlabel('Iterations');
ylabel('Mean Square Error (MSE)');
title('MSE Comparison');
legend('LMS','NLMS');

```

Outputs:





